

From Spreads to Spirals: How Financial Frictions Drive Lumpy Investments*

Miguel H. Ferreira[†] Timo Haber[‡] Hanbaek Lee[§]

April 28, 2026

Abstract

We show that firm-level real and financial frictions interact endogenously, resulting in a lumpy investment spiral: higher external financing costs delay investment, leading to lumpier investment, which in turn raises financing costs. This spiral generates endogenous state dependence in aggregate shock sensitivity: a deep downturn can endogenously arise even after mild negative aggregate real and financial shocks when firms' internal resources are scarce. We provide empirical evidence supporting the spiral mechanism.

Keywords : Investment, Financial Frictions, Firm Dynamics

JEL Codes: D53, E44, D21

*We thank our discussants Marcel Peruffo and John Finlay. For insightful comments and discussions we thank, Isaac Baley, Vasco M. Carvalho, Sara Casella, Julieta Caunedo, Andres Erosa, Miguel Faria-e-Castro, Alexandre Gaillard, Priit Jeenas, Joachim Jungherr, Julian Kozlowski, Tim Lee, Ramon Marimon, Davide Melcangi, Pascal Paul, Alessandra Peter, Ricardo Reis, Edouard Schaal, Tom Schmitz, Sanjay R. Singh, Vincent Sterk, Christian Wolf. For questions and comments, we also thank seminar participants at U.C.Davis, U.Kent U.Manchester, QMUL SEF, the SED (Cartagena), the EEA (Barcelona), Lisbon Macro Workshop, LuBraMacro (Porto), the PEJ (Braga), the MMF (Portsmouth), CICM (Hong Kong), International Conference on European Studies (Zurich), Midwest Macro (Kansas City), T2M (Paris).

[†]Queen Mary University of London. miguelthferreira@gmail.com

[‡]De Nederlandsche Bank, Eurosystem. habertimo@gmail.com or t.f.haber@dnb.nl. Views expressed are those of the authors and do not necessarily reflect official positions of De Nederlandsche Bank or the Eurosystem.

[§]University of Cambridge. hanbaeklee1@gmail.com

1 Introduction

Why do economies sometimes experience deep and protracted boom–bust cycles, such as the 2008 financial crisis, while at other times they undergo sharp recessions followed by rapid recoveries, as in the COVID-19 downturn? While differences in shock characteristics matter, understanding why shocks of comparable magnitude produce dramatically different dynamics remains central to macroeconomics. We argue that firms’ balance-sheet positions — specifically, their internal resources relative to investment needs — constitute a critical state variable that shapes aggregate dynamics, and show that in a full business-cycle environment this gives rise to a powerful state dependence.

This paper makes three contributions. First, we uncover and formalize a lumpy investment spiral at the micro level: a feedback loop between extensive-margin investment decisions and external financing costs. This spiral generates long inaction spells, large investment spikes, and tightly synchronized investment–debt issuance, all supported by firm-level data. Second, we show the spiral’s aggregate intensity is state-dependent. When firms enter downturns with low internal resources, more firms become trapped in the spiral. This amplifies investment delays and produces highly asymmetric aggregate dynamics in response to shocks of comparable magnitude. Third, financial shocks exhibit even stronger state dependence than TFP shocks. Even small financial shocks can endogenously generate disastrous recessions when hitting resource-scarce economies. This state-dependent amplification imposes substantial welfare costs, with consumption’s ergodic mean falling below its steady-state level.

We start the paper by using annual Compustat data on U.S. publicly listed firms, and documenting two robust empirical patterns. First, the extensive margins of debt issuance and investment move together over time: the fraction of firms with debt spikes consistently tracks the fraction with investment spikes. Second, investment lumpiness is positively associated with tighter financial constraints in the cross section. Firms with lumpier investment profiles tend to have higher leverage, lower distance to default, and greater reliance on external finance. These facts suggest a close link between investment and financing decisions, and motivate the central hypothesis of this paper: large investment events are associated with higher financing costs.

To better understand the mechanisms behind these empirical patterns, we develop a two-period model that incorporates both fixed adjustment costs and costly external finance. In this framework, investment spikes occur when firms exceed an inaction threshold, incurring both real and financial adjustment costs. Crucially, the model reveals that financial frictions alone can generate investment inaction, even in the absence of real ad-

justment costs. This occurs when there is a discontinuity between the internal and external cost of funds: if the marginal benefit of an additional unit of investment lies between the marginal cost of internal and external financing, a firm with insufficient internal resources to expand its capital stock will fall in an inaction region.

Moreover, the interaction between real and financial frictions gives rise to a self-reinforcing mechanism that we term the lumpy investment spiral. A firm that relies heavily on external financing may find it unprofitable to pay the adjustment cost and thus postpone investment. As investment is delayed, internal resources depreciate, increasing the cost of external finance needed to reach the same capital level. The higher financing cost then feeds back into both the extensive and intensive investment margins, completing the feedback loop.

We embed this mechanism in a business-cycle general equilibrium model with heterogeneous firms that face idiosyncratic productivity shocks, capital adjustment frictions, and costly external finance. Firms choose capital and external finance optimally, facing fixed costs of adjusting capital and a spread on external borrowing. The model is solved numerically and calibrated to match key U.S. firm-level moments, including the frequency and size of investment spikes, the average external financing share, and labor input. Importantly, the model also performs well on untargeted dimensions, such as the correlation between investment and debt issuance spikes, reproducing a correlation of 0.55 compared to 0.75 in the data.

The calibrated model reveals that the lumpy investment spiral survives aggregation and shapes macroeconomic dynamics in meaningful ways. Financial frictions lengthen inaction spells and magnify investment spikes, amplifying investment lumpiness across firms. These micro nonlinearities give rise to aggregate state dependence: the drop of investment and output in response to a negative TFP shock is, respectively, nearly 50% and 10% larger and more persistent when firms have low internal resources, as more firms fall into the spiral and delay investment.

The nonlinear aggregate effects stem primarily from high-productivity firms. These firms have the highest marginal benefit of investment and rely most on external finance. Their marginal benefit curve is also the most responsive to negative TFP shocks—an effect that is further amplified when internal resources are scarce. In addition, in low internal resources states, a larger mass of firms is closer to inaction and spiral regions. A negative productivity shock then compresses profits and pushes even more firms into the spiral, causing recessions to be deeper and more persistent. Shutting down either financial or real frictions eliminates this state dependence, as micro-level nonlinearities are no longer strong enough to survive aggregation.

Following, we extend the model to include an exogenous financial shock that affects the gap between internal and external financing costs. The addition of this shock gives rise to endogenous disaster risk: when a negative financial shock hits in a state of low internal resources, as it directly widens the gap between the cost of internal and external resources, it pushes a large mass of firms into the spiral, triggering exceptionally deep recessions. Over the business cycle, the presence of such endogenous disasters reduces the ergodic mean of consumption below its steady-state value, implying sizable welfare losses from nonlinear amplification. These results highlight the importance of jointly modeling real and financial frictions to understand the macroeconomic consequences of firm-level investment dynamics.

Lastly, we use firm-level Compustat data to provide empirical support for the spiral mechanism. First, we show that increases in external financing costs – instrumented by monetary policy shocks and industry level elasticities to the shock – lead to more lumpy investment: the probability of an investment spike declines, and the duration of inaction periods increases. Second, we find that investment spikes are followed by increases in proxies for interest rate spreads, consistent with the notion that large investment events raise firms’ external financing costs. Together, these findings provide evidence in support of both directions of the feedback loop central to the lumpy investment spiral.

Literature Empirical evidence on lumpy investment, such as that provided by [Cooper et al. \(1999\)](#) and [Cooper & Haltiwanger \(2006\)](#), documents the prevalence of lumpy investment, characterized by irregular, large-scale capital expenditures. These studies use plant-level data to demonstrate the presence of lumpy investment and its significant impact on aggregate investment volatility. Further, [Whited \(2006\)](#) demonstrates the significance of financial frictions for investment lumpiness using hazard rate estimation, although their focus is exclusively on a subsample of firms in Compustat. In contrast, our analysis utilizes all available data from 1974 to 2019. Our dataset also includes richer data on financial variables and lumpiness both in the cross-section and the panel, allowing for a more detailed analysis of how financial conditions influence investment behavior.

Building on this empirical foundation, there is a large literature on explaining lumpy investment dynamics in quantitative models. [Caballero & Engel \(1999\)](#), [Thomas \(2002\)](#), [Khan & Thomas \(2008\)](#), [Gourio & Kashyap \(2007\)](#), [Bachmann et al. \(2013\)](#), [Winberry \(2021\)](#), [Baley & Blanco \(2021, 2024\)](#), [Lee \(2022\)](#) and [Gnewuch & Zhang \(2025\)](#) study the effect of lumpy investment dynamics at the microeconomic level on aggregate outcomes. The focus in these papers is mostly on the real side, studying the importance of adjustment costs. Our model extends this literature by incorporating the financial side, where

firms cost of debt is increasing on the external financing needs. This addition provides a more comprehensive view of investment dynamics, including the importance of financing conditions in influencing firms' lumpy investment decisions.

Conversely, our work is also related to the literature on firm financing conditions and investment decisions. Studies by Kiyotaki & Moore (1997), Bernanke et al. (1999), Hennessey & Whited (2007), Jermann & Quadrini (2012), Khan & Thomas (2013), Senga et al. (2017), Begenau & Salomao (2018), Ottonello & Winberry (2020), Ferreira et al. (2023), Jeenas (2023) and Cloyne et al. (2023) have explored how financial frictions impact investment behavior. These studies illustrate that financial constraints can significantly affect the timing and magnitude of investment, thereby influencing aggregate economic outcomes. Jiao & Zhang (2022) depart from this literature and closely examine the interaction between real and financial frictions, although they do not address aggregate implications in depth. In a similar vein, Melcangi (2024) focuses on the interplay between financial and real frictions, particularly in the context of employment, providing valuable insights into how financing conditions affect labor market dynamics. Additionally, Bazdresch (2013) show that financing costs are crucial for explaining investment dynamics but do not dissect the interaction in a detailed model. We add to this literature by incorporating investment lumpiness into a financing frictions model, highlighting the combined effects of financial constraints on lumpy investment behavior. Likewise, Koby & Wolf (2020) present a model with collateral constraints and financing frictions, studying their effects on investment decisions. However, we further advance this by exploring the possible non-linearities over the business cycle that these two frictions can generate.

Outlook The rest of the paper is structured as follows. Section 2 presents the stylized facts. Section 3 describes the basic theory. This is followed by Section 4, which describes the model before we present our results thereof in Section 5. Section 6 presents empirical support for the spiral mechanism. Finally, Section 7 concludes.

2 Stylized facts

In this section, we use annual firm-level data from Compustat covering publicly listed U.S. firms between 1974 and 2024 to establish two key stylized facts. As is known, Compustat provides a long panel of firms' capital expenditures, balance sheet items, and financing costs, which allows us to measure both investment outcomes and financial positions in a consistent way across time. To our knowledge, Compustat is the only U.S.

dataset with the necessary length and richness to document firm-level patterns of investment lumpiness and its relation to financial frictions.

2.1 Data description

Our main outcome of interest is firm investment. Following [Ottonello & Winberry \(2024\)](#), we define the investment rate as the ratio of capital expenditures (Compustat code CAPX) to the lagged value of property, plant, and equipment (code PPEGT). In line with the literature, we define a large or lumpy investment episode whenever the investment rate exceeds 20% (see e.g. [Bachmann & Ma, 2016](#); [Cooper & Haltiwanger, 2006](#)).

In addition to investment, we utilize several firm observables that characterize financial positions. These include (net) debt issuance rate, leverage, liquidity, and interest expenses, constructed using standard definitions from the corporate finance literature. A key measure is distance to default, which we use as a proxy for the external finance premium as it has been shown to be negatively and significant predictor for actual bond spreads (see e.g. [Ferreira et al., 2024](#); [Gilchrist & Zakrajšek, 2012](#)). Distance to default is constructed using CRSP stock price data and the methodology of [Merton \(1974\)](#) and [Gilchrist & Zakrajšek \(2012\)](#). Appendix A.1 provides further details on the data cleaning procedure and descriptive statistics.

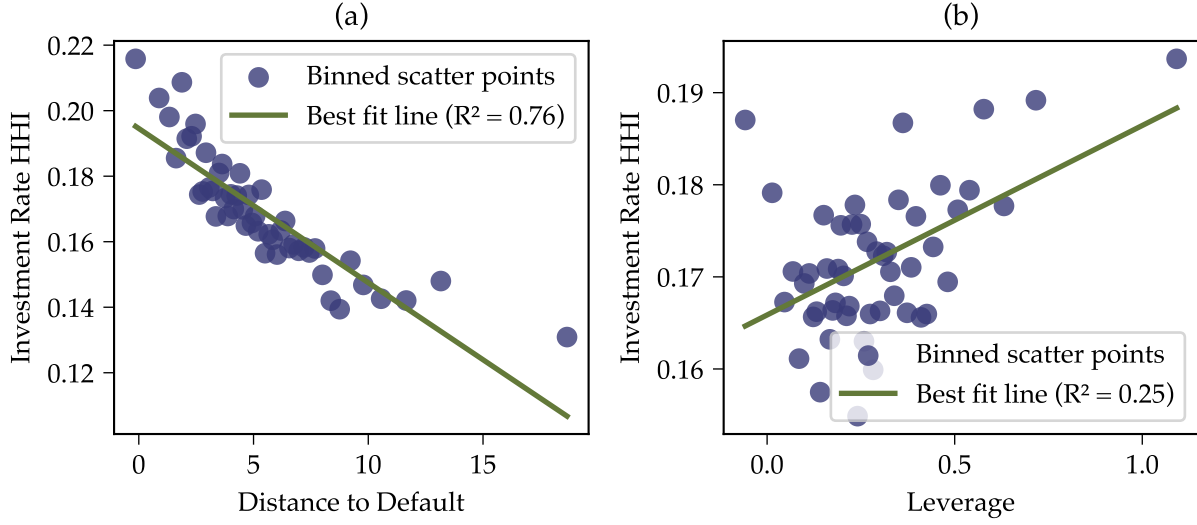
To capture investment lumpiness at the firm-level, we construct measures of how concentrated firm-level investment is over time. To ensure sufficient time-series variation, we restrict the sample to firms with at least ten years of data. Our main measure of lumpiness is the Herfindahl–Hirschman Index (HHI) of concentration, calculated at the within firm level:

$$\text{HHI}_j = \sum_{t=1}^{T_j} \left(\frac{i_{jt}}{\sum_{s=1}^{T_j} i_{js}} \right)^2, \quad \text{where } i_{jt} > 0. \quad (1)$$

where i_{jt} denotes firm j 's investment rate i and T_j the number of periods the firm j is observed. We restrict attention to periods of positive investment rates so that the HHI measures the concentration of active investment episodes. Higher values of this index indicate greater lumpiness, i.e., a concentration of investment in fewer periods.

For robustness, we also construct multiple alternative measures, including Appendix A.3 provides further details on these definitions and their properties.

Figure 1: Investment concentration and financial position



Note: Binned histograms between distance to default and HHI in investment on the LHS. On the RHS the x-axis measures leverage instead. In both cases there are 50 bins (dark blue squares) and the regression line (solid orange line). We control for various observables as illustrated in Equation (2)

2.2 Stylized Fact 1: Increased lumpiness is associated with tighter financial constraints

We first document the strong positive correlation between investment lumpiness and the financial state of firms. We test this correlation by running cross-sectional regressions in the following fashion.

$$\text{Lumpiness}_j = \alpha + \sum_{s=1}^S \gamma_s \mathbf{1}[j \in s] + \beta \text{Average Financial Position}_j + \Gamma X_j + \epsilon_j, \quad (2)$$

where γ_s is the sector specific intercept for firm in major industry s , β is the correlation coefficient of interest and X_j is a set of firm controls comprised of average real sales growth, average real total assets, average share of current assets, average age and average 5-year standard deviation of real sales growth.

Figure 1 displays the graphical evidence of this correlation, plotting binned scatter plots between the average distance to default (panel (a)) and leverage of a firm (panel (b)) and the residualized Herfindahl-Hirschman index of investment rate of that firm. The solid line depict the regression line (i.e. β) whilst the scatter points are binned on a firm level with 50 total bins. It is clear that lumpiness is negatively correlated with distance to default and positively with leverage. We interpret this relationship as reduced-form

evidence that firms with more lumpy investment behavior tend to operate closer to their default boundary, consistent with greater financial fragility and, consequently, higher implied credit spreads.¹ The pattern suggests that firms whose capital adjustment is concentrated in a small number of large spikes systematically face tighter financial conditions.

This evidence should not be interpreted causally. Investment lumpiness may be correlated with unobserved firm characteristics. For example, we do not control for financing constraints, growth opportunities, or sectoral cyclicalities which may drive both investment timing and financial fragility. The results therefore represent descriptive correlations rather than structural identification, potentially reflecting deeper underlying mechanisms. Nevertheless, the strong association between investment lumpiness and proximity to default provides informative empirical discipline for the mechanism in our model and motivates the subsequent analysis of how non-convex investment frictions interact with financial constraints to amplify leverage and spread dynamics.

We also examine correlation coefficients using alternative proxies for external finance dependence, including the liquidity ratio and interest expenses. The results remain robust across these different measures of financial position. In addition, we repeat the analysis using several alternative measures of investment lumpiness at the firm level. The full set of results is reported in Table A2 in Appendix A.3.

2.3 Stylized fact 2: Investment and debt spikes co-move.

Second, we document a strong co-movement between investment and debt spikes.² Figure 2 plots the percentage of firms with an investment rate (blue line) and net debt issuance rate (orange line) above 20% in a given year. It is noticeable the strong co-movement between the two series. Periods with a higher fraction of large investment are also characterized with by a higher fraction of firms with large net debt issuance rates.

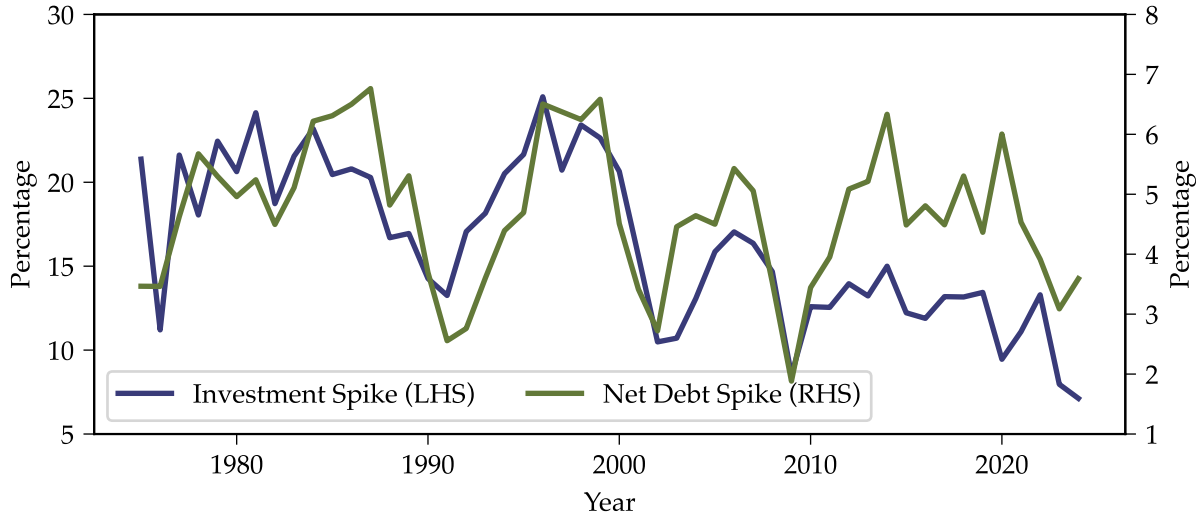
In the Appendix we further confirm that this fact is not an artifact of the threshold chosen for large investment or debt issuance episodes. We do so by varying the spike classification cutoff between 30% and 10%. Figure A2 in the Appendix shows that the time-series correlation between the average share of firms with net debt spikes and investment spikes is robust to using these alternative cutoffs.

The two stylized facts presented in this section highlight that lumpy investment and debt decisions are correlated and that the decision of large investments is inextricably linked to (the change of) the external finance position. More specifically, the evidence

¹Distance-to-default is widely used as a market-based proxy for credit spreads; see Gilchrist & Zakrajšek (2012) and Ottonello & Winberry (2020).

²A debt spike is defined as net debt issuance higher or equal than 20% of total liabilities.

Figure 2: Percent of firms with investment and net debt issuance spikes over time



Note: This figure plots both the percent of firms with a investment spike and with a net debt issuance spike between 1975 and 2018. The left axis measures the percent of firms with an investment spike whilst the right axis measures the percent of firms with a net debt issuance spike.

suggests that firms with a lumpier investment profile are more risky, which should result in higher spreads. To understand the mechanisms that connect the two decisions, as well as its implications for macro outcomes, we build a firm's investment model, for which we can derive theoretical predictions, that is introduced in the next section.

3 Theory and illustration

This section develops a simple firm investment model to rationalize the close link between investment lumpiness and financial conditions documented in the previous section. The model establishes that financial frictions alone can generate investment inaction. We further demonstrate how these frictions amplify real adjustment costs, producing what we term *lumpy investment spirals*. To highlight the mechanism as transparently as possible, we deliberately adopt a parsimonious framework that nests financial frictions in their simplest form. Specifically, we assume that external financing is more costly than internal financing, thereby introducing a wedge that captures the essence of financial constraints without requiring additional structure on the source of the friction.

3.1 Model Setup

Consider a representative firm with initial capital k , which depreciates at rate δ , and productivity z . Output is produced using a Cobb–Douglas technology with decreasing returns to scale:

$$\psi(k, z) = zk^\alpha + (1 - \delta)k. \quad (3)$$

The firm uses its profits $\psi(k, z)$ to reinvest in capital, for production in the second period. There is no uncertainty, and so productivity tomorrow z' is known at the time of investment. After the second period the firm exits and distributes profits as dividends. The firm chooses next-period capital k' to solve

$$\max_{k'} \psi(k, z) - \mathcal{R}(k', k, z)k' + \beta\psi(k', z'), \quad (4)$$

where $\mathcal{R}(k', k, z)$ is the average cost of finance, defined as

$$\mathcal{R}(k', k, z) = \begin{cases} 1 & \text{if } k' \leq \psi(k, z), \\ \frac{\psi(k, z)}{k'} + \mathcal{R}^{ext} \left(1 - \frac{\psi(k, z)}{k'}\right) & \text{if } k' > \psi(k, z). \end{cases} \quad (5)$$

Expression (5) implies that, if investment can be fully financed with internal funds, the average (and marginal) cost equals one. However, if the firm chooses k' such that external financing is required, the average cost rises with the share of external funding. The marginal external financing cost $\mathcal{R}^{ext}$ is parameterized as:

$$\mathcal{R}^{ext} = 1 + e_0 + e_1 (k' - \psi(k, z)), \quad (6)$$

where e_0 is the baseline spread and e_1 captures the sensitivity of costs to financing needs. This sensitivity could be interpreted as higher dependence on external financing leading to an increase in risk, similar to the micro-foundation provided in [Ottonello & Winberry \(2020\)](#).³ $\mathcal{R}^{ext}$ overall can be interpreted as the spread a firm will pay in external financing, which could be in the form of loan, bond or equity issuance.⁴

Finally, the specific form of $\mathcal{R}^{ext}$ nests the equity issuance cost of [Hennessy & Whited \(2007\)](#).

³The external finance structure is consistent with micro-foundations such as [Ottonello & Winberry \(2020\)](#). Proposition A.2 and Remark 1 in Appendix B.1 show how e_0 and e_1 can be mapped to a micro-founded spread.

⁴The baseline model most closely resembles equity issuance. Proposition A.1 in Appendix B.1 shows how the formulation maps into a debt financing case with a small scale adjustment to $\mathcal{R}^{ext}$.

Remark 1 (External finance costs equivalent to equity financing cost).

$\mathcal{R}^{ext}$ maps into the equity issuance costs in *Hennessy & Whited (2007)*, with e_0 and e_1 corresponding to λ_1 and λ_2 in *Hennessy & Whited (2007)*.

Relative to *Hennessy & Whited (2007)*, the key difference is that we impose this structure directly on the marginal cost of external finance, which could be in the form of loans, bonds or equities. This ensures the value function remains strictly concave, as the discontinuity is present in the marginal cost, eliminating the need for tie-breaking rules. Moreover, e_0 and e_1 could be directly estimated in loan-level data, where they would be interpreted as spreads.

3.2 External financing costs can result in investment inaction

We now proceed to illustrate how the financial friction can generate inaction in investment. Our finding is summarized in the next proposition:

Proposition 1 (Financial frictions generate investment inaction).

When $e_0 > 0$, the marginal cost curve exhibits a discontinuity that generates inaction periods, during which $k' \leq k$, if $\pi(k, z) \leq \delta k$.

Proof.

The firm's value function is given by:

$$V(k, z) = \max_{k'} \psi(k, z) - \mathcal{R}(k', k, z)k' + \beta\psi(k', z') \quad (7)$$

Taking the derivative with respect to k' , we obtain the marginal benefit (MB) and marginal cost (MC) curves:

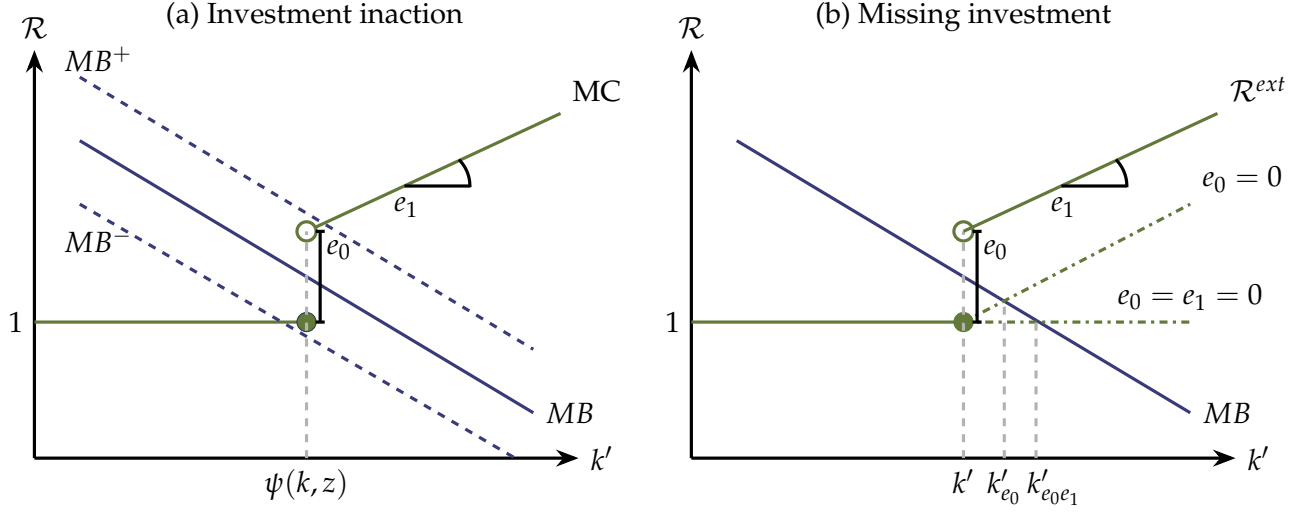
$$MB = \beta \frac{\partial \psi(k', z')}{\partial k'} \quad \text{and} \quad MC = \mathcal{R}(k', k, z) + k' \frac{\partial \mathcal{R}(k', k, z)}{\partial k'} \quad (8)$$

Thus, the marginal cost function exhibits a discontinuity at $k' = \psi(k, z)$, as:

$$\lim_{k' \rightarrow (\psi(k, z))^-} MC = 1 \quad \text{and} \quad \lim_{k' \rightarrow (\psi(k, z))^+} MC = 1 + e_0 \quad (9)$$

If $e_0 > 0$, this discontinuity creates regions where $MB \neq MC$ leading to firms delaying investment. When a firm's profits do not suffice to cover the capital depreciation— $\pi(k, z) \leq \delta k$ —firms fall into inaction, in which case $k' \leq k$. $\square$

Figure 3: Marginal benefit and marginal cost of investment



Note: This figure plots both the marginal cost and marginal benefit curves as a function of capital. Note the jump in the marginal cost curve occurs at the point when the firm exhausts all internal resources and to further invest has to use external resources. Left panel illustrates the inaction region caused by a discontinuity in the marginal cost curve. Right panel illustrates the missing investment caused by e_0 and e_1 .

An illustration of Proposition 1 is provided in Figure 3a. The figure depicts the case in which the marginal benefit curve lies entirely within the discontinuity of the marginal cost schedule and does not intersect it. In this situation, the firm uses its internal resources up to $\psi(k, z)$ but refrains from seeking external finance, since the marginal benefit never crosses the external financing cost. Even small shifts in the marginal benefit curve within this range do not translate into changes in investment. Whether the firm is in an inaction region depends on profitability relative to depreciation: if internal profits exceed δk , the firm can still accumulate some capital even without accessing external finance; if profits are low, the firm cannot raise capital above its initial level and is in an inaction region. So, the firm's profitability together with capital depreciation rate are key objects to determine if the firm is in an inaction region or not.

If, instead, we consider the case when the marginal benefit curve would intersect the schedule above $k' = \psi(k, z)$, as indicated by the top dashed line, the firm would combine internal funds with external financing in the amount $k' - \psi(k, z)$, paying the external rate $\mathcal{R}^{ext}$. Alternatively, if the marginal benefit curve would intersect the marginal cost schedule below $k' = \psi(k, z)$, as indicated by the bottom dashed line, internal funds would suffice to finance the optimal investment.

Figure 3b illustrates the central role of e_0 and e_1 in shaping both the extent of inaction and the aggregate investment shortfall generated by financial frictions. The fraction

of firms that remain inactive, as well as the length of the inaction region, naturally depends on the size of e_0 . When $e_0 = 0$, the marginal cost curve is continuous with only a kink, and no inaction arises. Conversely, the larger is e_0 , the stronger the discontinuity in the marginal cost curve and the greater the fraction of firms that remain inactive. Since inaction occurs only when internal profits are insufficient to finance desired capital accumulation, it predominantly affects less productive firms.

The same figure also illustrates the investment shortfall induced by these frictions. When $e_0 = 0$, the marginal cost curve exhibits no discontinuity, and tomorrow's capital rises from k' to k'_{e_0} . The gap between these two levels represents the portion of capital lost due to the discontinuity. If, in addition, e_1 is set to zero, capital increases further to $k'_{e_0 e_1}$, capturing the additional shortfall in capital attributable to financial frictions.

3.3 Lumpy investment spirals

In the previous subsection we establish that, as long as there is a discontinuity in the marginal cost curve when the firm access external financing, this will generate investment inaction. We now prove that, besides financial frictions directly generating investment inaction, they also amplify the effects of a real friction in the form of a fixed capital adjustment cost. To illustrate this, consider now the model in the previous section with the addition of a capital adjustment cost ξ and, for simplicity, we assume $k' \geq (1 - \delta)k$, this is, firms do not disinvest. If the firm pays the cost and invests, its value is

$$V^A(k, z) = \max_{k' > (1-\delta)k} \psi(k, z) - \mathcal{R}(k' + \xi, k, z)(k' + \xi) + \beta\psi(k', z'), \quad (10)$$

and the value of the firm choosing not to pay the adjustment cost, thereby precluding it from adjusting the capital stock, is:

$$\begin{aligned} V^{NA}(k, z) &= \max_{k'} \psi(k, z) - \mathcal{R}(k', k, z)k' + \beta\psi(k', z'), \\ \text{subject to: } &k' = (1 - \delta)k. \end{aligned} \quad (11)$$

The firm will decide to pay the adjustment cost if the value under investment is higher than the value under no adjustment. This decision will be affected by $\mathcal{R}$, as established in the following proposition:

Proposition 2 (External finance costs elevate investment inactivity).

A firm makes an investment following the cutoff rule with respect to productivity:

$$V = \begin{cases} V^A & \text{if } z' > z^*, \\ V^{NA} & \text{if } z' \leq z^*. \end{cases} \quad (12)$$

As the cost of external finance increases, the threshold rule z^* increases:

$$\frac{\partial}{\partial \mathcal{R}} z^*(k; \mathcal{R}) > 0. \quad (13)$$

Proof. A firm invests if:

$$V^A(k, z) > V^{NA}(k, z) \quad (14)$$

Higher external finance costs $\mathcal{R}$ increases the cost of investing:

$$\frac{\partial V^A}{\partial \mathcal{R}} < 0 \quad (15)$$

As replacement investment never requires external finance, V^{NA} does not vary with e_0 and e_1 , and so the threshold z^* must increase:

$$\frac{\partial z^*}{\partial \mathcal{R}} > 0 \quad (16)$$

Thus, higher financing costs reduce the probability of investment.

□

Proposition 2 establishes how the cost of external finance shapes the firm's investment decision on the extensive margin. While we characterize the overall impact of external financing costs on the decision to pay the fixed adjustment cost and invest, both e_0 and e_1 play distinct roles in driving this result. The fixed component e_0 , as shown earlier, can by itself generate inaction. Even when capital adjustment costs are low, a sufficiently high e_0 may induce the firm to forgo investment altogether, all else equal. This translates into a higher productivity threshold z^* to trigger investment. By contrast, e_1 on its own does not create inaction in the same way as e_0 . Instead, as e_1 increases, the firm reduces investment along the intensive margin. This lower level of investment may render it unprofitable to pay the fixed adjustment cost, effectively lengthening the inaction region through the interaction between e_1 and capital adjustment costs. As a result, increases in e_1 also raise the productivity threshold z^* .

Finally, we turn to how higher adjustment costs amplify the effects of external financ-

ing costs:

Proposition 3 (The monotonicity of the bond interest rate in the capital adjustment cost).

Suppose the capital adjustment cost ξ is expressed as $\xi = \tilde{\xi} + \kappa$, where κ is non-stochastic and $\tilde{\xi}$ is the stochastic part of the adjustment cost. The bond interest rate increases in the liquidation value shifter κ :

$$\frac{\partial}{\partial \kappa} \mathcal{R} \geq 0 \quad (17)$$

Proof.

The firm's borrowing function is:

$$\mathcal{R} = \mathcal{R}(k', k', z) \quad (18)$$

A higher adjustment cost ξ increases financing needs:

$$\frac{\partial k' + \xi}{\partial \kappa} > 0 \quad (19)$$

Since $\mathcal{R}$ is an weakly increasing function of $k' + \xi$:

$$\frac{\partial \mathcal{R}}{\partial \kappa} \geq 0 \quad (20)$$

Thus, a higher ξ raises external financing cost.

□

Proposition 3 establishes that a larger adjustment cost, which leads to larger external financing needs all else equal, causes the external financing cost to increase. This channel operates more through e_1 as an increase in external finance leads to a higher marginal cost via e_1 and is independent of e_0 .

The impact that an increase in lumpiness has on external financing cost and the feedback effect that financing costs have on the investment lumpiness gives rise to what we term lumpy investment spirals.

In the next section we move to extend the current setting into a multi-period, general equilibrium model. In that setup the mechanism will work as follows. In period one, the firm starts with a given probability of investment ψ^* and a given financing cost $\mathcal{R}$, which depends on the investment level. If the firm does not incur in large-scale investment in period 1, it will enter period two further away from the optimal capital level due to capital depreciation. As such, the optimal investment level is higher, and the firm becomes more

lumpy. This translates into a higher financing cost, with $\mathcal{R}' > \mathcal{R}$ (Proposition 3), which feeds back to the investment threshold rule, with $z^{*'} > z^*$ (Proposition 2). If the firm does not invest in period 2, the situation it will face in period three will be even worse. This continues until the firm incurs a large-scale investment profile and gets out of the spiral.

4 Structural model

We now describe a business cycle, general equilibrium model with heterogeneous firms to quantify the importance of the investment spiral mechanism found with the theoretical model. Time is discrete and infinite. There is a fixed mass of heterogeneous firms $j \in [0, 1]$ that face variations in productivity and are subject to financial frictions and real frictions. The household is modeled as a simple representative agent who works, saves and consumes.

4.1 Firms

At the beginning of each period, a firm's state is given by its idiosyncratic productivity z and its predetermined capital stock k . The firm is also aware of the distribution Φ of the individual states (z, k) and aggregate productivity A , rationally expecting the dynamics of these aggregate allocations. Define $S \equiv \{\Phi, A\}$ as the collection of aggregate states at time t in order to ease on notation.

Production Each firm operates using the following Cobb-Douglas production function, where the inputs are labor n and the capital k :

$$y = Azk^\alpha n^\gamma \quad (21)$$

with $\alpha > 0$, $\gamma > 0$ and $\alpha + \gamma < 1$. Both the idiosyncratic and the aggregate productivity shock follow a log-AR(1) process:

$$\log(z') = \rho_z \log(z) + \sigma_z \epsilon'_z, \quad \text{where } \epsilon'_z \sim N(0, 1). \quad (22)$$

$$\log(A') = \rho_A \log(A) + \sigma_A \epsilon'_A, \quad \text{where } \epsilon'_A \sim N(0, 1). \quad (23)$$

where $0 \leq \rho_z < 1$, $0 \leq \rho_A < 1$, $\sigma_z > 0$ and $\sigma_A > 0$.

Based on this production technology, the optimal labor demand $n(k, z)$ and production

volume $y(k, z)$ is determined from the following static maximization problem:

$$\max_n Azk^\alpha n^\gamma - w(S)n \quad (24)$$

where w denotes the aggregate wage, which depends on the aggregate state S . This optimization results in the following expressions:

$$n(k, z, S) = \left(\frac{\gamma Azk^\alpha}{w(S)} \right)^{\frac{1}{1-\gamma}} \quad (25)$$

$$y(k, z, S) = (Az)^{\frac{1}{1-\gamma}} \left(\frac{\gamma}{w(S)} \right)^{\frac{\gamma}{1-\gamma}} k^{\frac{\alpha}{1-\gamma}} \quad (26)$$

and, finally, the static profit of a firm conditional on optimal labour demand can be expressed as:

$$\pi(k, z, S) = (1 - \gamma)y(k, z, S). \quad (27)$$

Real constraints Each firm owns the capital stock and makes an investment decision. Firm investment is subject to a fixed capital adjustment cost friction. In particular, the firm incurs the fixed adjustment cost $\xi \sim_{iid} \mathcal{U}[0, \bar{\xi}]$ when investment is beyond a range Ω that is proportional to the existing capital stock:

$$\Omega = [(1 - \delta - \nu)k, (1 - \delta + \nu)k]. \quad (28)$$

4.2 Financial intermediaries

To finance investment the firm can use either internal resources or external resources. If the firm uses external finance, the cost incurred follows the same form as in Section 3,

$$\mathcal{R}^{ext} = 1 + e_0 + e_1 (k' - \pi(k, z, S) - (1 - \delta)k). \quad (29)$$

Again, e_0 can be interpreted as the average baseline cost on borrowing or equity issuance, and e_1 captures the effect that higher external financing needs are going to lead to higher marginal costs, which could be explained by a higher increase in risk, similar to the micro-foundation provided in [Ottonello & Winberry \(2024\)](#). $\mathcal{R}^{ext}$ overall can be interpreted as the spread a firm pays if using external financing, which could be in the form of loan, bond or equity issuance. Defining $\psi(k, z; S) = \pi(k, z; S) + (1 - \delta)k$, the overall investment

finance cost is

$$\mathcal{R}(k', k, z; S) = \begin{cases} 1 & \text{if } k' \leq \psi(k, z; S), \\ \frac{\psi(k, z; S)}{k'} + \mathcal{R}^{ext}(k', k, z; S) \left(1 - \frac{\psi(k, z; S)}{k'}\right) & \text{if } k' > \psi(k, z; S). \end{cases} \quad (30)$$

If investment can be fully financed with internal funds, the marginal cost equals one. If external financing is required, the marginal cost rises with the share of external funding.

4.3 Firm's problem

As previously outlined, a firm starts the period with a given idiosyncratic productivity z shock, and capital stock k . The firm decides if and how much to invest i and dividends to distribute d . The value of a firm is given by

$$V(k, z, S) = \psi(k, z, S) + \int_0^{\bar{\xi}} \max \{V^a(k, z, \xi, S), V^{na}(k, z, S)\} dG(\xi), \quad (31)$$

where V^a is the value if the firm pays the adjustment cost and decides to invest, while V^{na} is the value under no adjustment. V^a is defined as follows:

$$V^A(k, z, \xi, S) = \max_{k'} \{-\mathcal{R}(k', k, z, S)(k' + \xi w(S)) + \mathbb{E}[\Lambda(S, S')V(k', z', S')]\},$$

When the firm decides to adjust the capital stock, it can freely choose k' , but it has to pay the adjustment cost ξ , denoted in labour units. The firm also decides the amount of external finance, on which it pays the cost $\mathcal{R}^{ext}$.⁵ $V(z', k', S')$ captures the continuation value of the firm.

The value under no adjustment V^{na} is:

$$V^{NA}(k, z, S) = \max_{k^c \in \Omega(k)} \{-\mathcal{R}(k^c, k, z, S)k^c + \mathbb{E}[\Lambda(S, S')V(k^c, z', S')]\}. \quad (32)$$

The main difference relative to the adjustment case is that $k^c \in \Omega(k)$ and the firm does not incur the capital adjustment cost.⁶

⁵In the baseline model, the fixed adjustment cost affects external financing costs indirectly through the firm's total costs and its investment decision, whereas the theory in Section 3 abstracts from this by imposing an exogenous direct effect on external financing costs. This choice is intended to facilitate efficient computation in the baseline model and theoretical clarity in the simple model, without compromising the equivalence of the economic mechanism.

⁶Although firms are theoretically able to use external finance without adjusting their capital stock, under our calibration—particularly due to the choice of ν —this option is never exercised.

4.4 Household

We close the model by introducing a representative household, who consumes, saves, and supplies labor. The household specification closely follows [Khan & Thomas \(2008\)](#) and [Bachmann et al. \(2013\)](#). Specifically, we assume a log utility and disutility for indivisible labor supply in the following form:

$$U(C, L) = \log(C) - \eta L, \quad (33)$$

where C is consumption; L is the labor supply; η is the disutility parameter for labor supply.

The recursive formulation of the household's problem is as follows:

$$J(a) = \max_{C, a', L} \log(C) - \eta L + \beta \mathbb{E}J(a') \quad (34)$$

$$\text{s.t. } C + \Lambda(S, S')a' = w(S)L + a \quad (35)$$

where a is the state-contingent equity portfolio value; $\Lambda(S, S')$ is the stochastic discount factor for the equity portfolio; R^f is the risk-free interest rate.

From the first order condition with respect to state contingent saving a' , we characterize the discount factor as follows:

$$\Lambda(S, S') \equiv \beta \frac{C(S)}{C'(S')}. \quad (36)$$

As in [Khan & Thomas \(2008\)](#), we define $P := 1/C$, which we use for normalizing the firm's value function for easier computation.

4.5 Equilibrium

We are now able to define a recursive competitive equilibrium in this economy.

Definition 1 (Recursive Equilibrium). *An equilibrium of this model is a set of functions $V(k, z, S)$, $k'(k, z, S)$, $k^c(k, z, S)$, $\Phi(k, z, S)$, $\widehat{\Phi}(k, z, S)$, Λ , p , w , a and C that solve the firm, financial intermediary and household problem and also clear the market for labor, output and equity, as described by the following conditions.*

1. *Production firms optimization: $V(k, z, S)$ solves Equation (31) with associated decision rules $k'(k, z, S)$ and $k^c(k, z, S)$.*

2. The stochastic discount factor of the household is given by Equation (36). The wage satisfies the intratemporal optimality condition $w = \eta C$.
3. Aggregate investment is defined as $I = K - (1 - \delta)K$, where $K = \int k d\Phi_t(k, z, S)$. Aggregate consumption is defined by $C = Y - I$, where $Y = \int y d\Phi_t(k, z, S)$.
4. The goods market clears $Y = C + I$
5. The labor market clears $L = \int n d\Phi_t(k, z, S)$
6. Equity market clears by Walras Law.

4.6 Solving and calibrating the model

Solution Method We use the global sequence-space solution method developed in Lee (2025) — the repeated transition method (RTM). Because of the non-linearity on the firm level, aggregate fluctuations in the recursive competitive equilibrium are potentially highly nonlinear. Since our main focus is on endogenous state dependence in the propagation of exogenous shocks, we adopt this global nonlinear method, which recovers state-by-state transitions accurately without imposing local approximations. The approach exploits the ergodicity of the recursive competitive equilibrium: with a sufficiently long simulated path for the exogenous states, essentially all relevant combinations of endogenous and exogenous states are realized. Each period's conditional expectation of value functions is then computed by combining value functions from periods with tightly similar endogenous states but differing exogenous realizations. The method does not rely on a parametric law of motion. Instead, only a metric that captures the similarity of endogenous states across time is needed. Implementation details are provided in Appendix C.1.

Steady state calibration Each period in the model corresponds to one year. For most parameters, we follow Khan & Thomas (2008), with the full set of fixed parameters reported in Table C1 in Appendix C.1. The discount factor β is set to generate an average annual real interest rate of 2.3%. The production parameters, α and γ , imply a labor share of 64% and a capital share of 26%, respectively. Leisure preferences are calibrated so that households work roughly one third of their available time, and the depreciation rate is set at 9%. The mean productivity level is normalized such that, once transformed into a log-normal distribution, the average productivity equals one. The persistence and variance of the idiosyncratic productivity shock, ρ_z and σ_z , are set to 0.859 and 0.022, respectively, following Khan & Thomas (2008).

Table 1: Calibrated model fit

Moment	Data	Model	Reference
Targeted moments			
Mean investment spike rate (%)	14.4	10.0	Zwick & Mahon (2017)
Mean investment rate (%)	10.4	10.6	Zwick & Mahon (2017)
Positive investment rate (%)	85.6	90.0	Winberry (2021)
Std investment rate (%)	19.0	18.8	Zwick & Mahon (2017)
Mean external finance/capital (%)	34.0	27.0	OW20 & CM20
Mean external financing spread (%)	2.04	2.59	Ferreira et al. (2024)
Labor hours	0.33	0.34	8 working hours per day
Untargeted moments			
Correlation between investment and debt spikes	0.75	0.55	Compustat Annual

Note: We measure the correlation between the investment and debt spikes at the aggregate level. OW20 stands for [Ottonello & Winberry \(2020\)](#) and CM20 for [Crouzet & Mehrotra \(2020\)](#).

The remaining parameters which include the upper bound of the adjustment cost ξ , the investment replacement range ν , and the external financing cost parameters e_0 and e_1 are calibrated to match investment and external financing moments. For investment, we use moments reported by [Zwick & Mahon \(2017\)](#), whose dataset includes both large publicly listed and small private U.S. firms. For financial moments, we target the mean external finance-to-capital ratio reported by [Crouzet & Mehrotra \(2020\)](#), whose sample also includes small private firms; notably, this moment is similar to the one documented by [Ottonello & Winberry \(2020\)](#) for publicly listed firms. To discipline the average external financing cost, we rely on the estimates of firm-level spreads from [Ferreira et al. \(2024\)](#). The calibrated values of these parameters are shown in Table C2, while Table 1 presents the model’s fit to the targeted moments.

Beyond matching the targeted investment and financing moments, the model also reproduces key comovement patterns in the data. In particular, it predicts an aggregate correlation of 0.55 between investment and debt issuance spikes, compared to 0.75 in the data. Figure C1 in Appendix C.2—the model counterpart to Figure 2—illustrates this strong correlation between investment and external financing spikes.

5 Equilibrium Analysis

In this section, we investigate the macroeconomic implications of the lumpy investment spiral. We begin by showing how the interaction between financial and real frictions generates nonlinear business cycle dynamics, where the response of aggregate investment and output to TFP shocks depends endogenously on the economy's current state. We then examine how the presence of financial shocks can give rise to endogenous disaster risk.

5.1 Equilibrium analysis

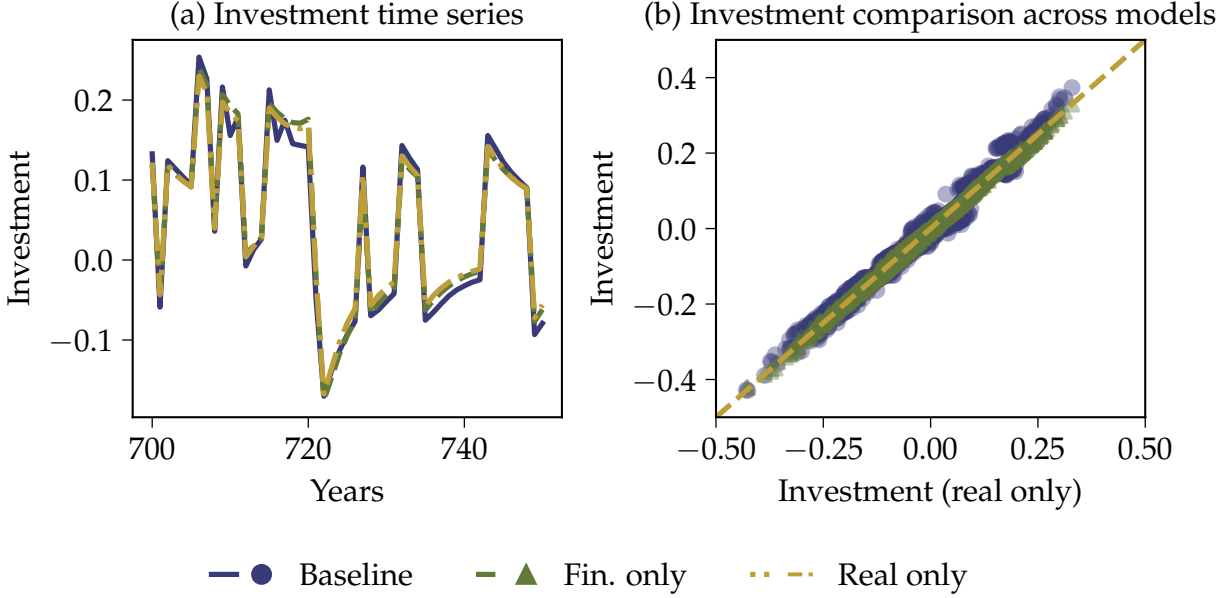
We begin by examining whether the investment spiral derived in Section 3 survives aggregation and remains relevant at the business-cycle level. The left panel of Figure 4 compares aggregate investment dynamics across three model variants: the baseline model (solid blue line), a model with only real frictions à la Khan & Thomas (2008) (orange dashed line), and a model with only financial frictions (brown dotted line). While the single-friction models produce nearly identical investment dynamics, the baseline model differs markedly. Moreover, investment dynamics in the single-friction models are well approximated by a linear fit, whereas the baseline model exhibits strong nonlinearities. This indicates that, under identical shocks, aggregate responses depend on the prevailing state of the economy.⁷

The right panel of Figure 4 further highlights this nonlinearity. It compares aggregate investment in the baseline model and in the financial-friction-only model to that in the real-friction-only model. If investment behavior were similar across models, the scatter plots would lie along the 45-degree line. Instead, the baseline model shows substantial dispersion, underscoring its nonlinear dynamics, while the financial-friction-only model aligns closely with the 45-degree line.

Given the micro-level differences in investment behavior generated by the two single-friction models—where, in the real-friction case, firms experience long inaction spells and invest in large bursts, whereas in the financial-friction model, constrained firms invest up to their borrowing limits while unconstrained firms equate the marginal product of capital to the cost of capital—it is striking that aggregate investment dynamics remain so similar across them. This illustrates the strength of general equilibrium forces, which largely wash out micro-level heterogeneity when frictions act in isolation. In contrast, when both real and financial frictions coexist in the baseline model, general equilibrium forces no longer neutralize micro-level differences. Their interaction amplifies firm-level

⁷Figure C2 in Appendix C.4 reinforces this point by comparing the laws of motion of aggregate capital across the three model variants, each relative to a linear fit.

Figure 4: Comparison of investment dynamics across models

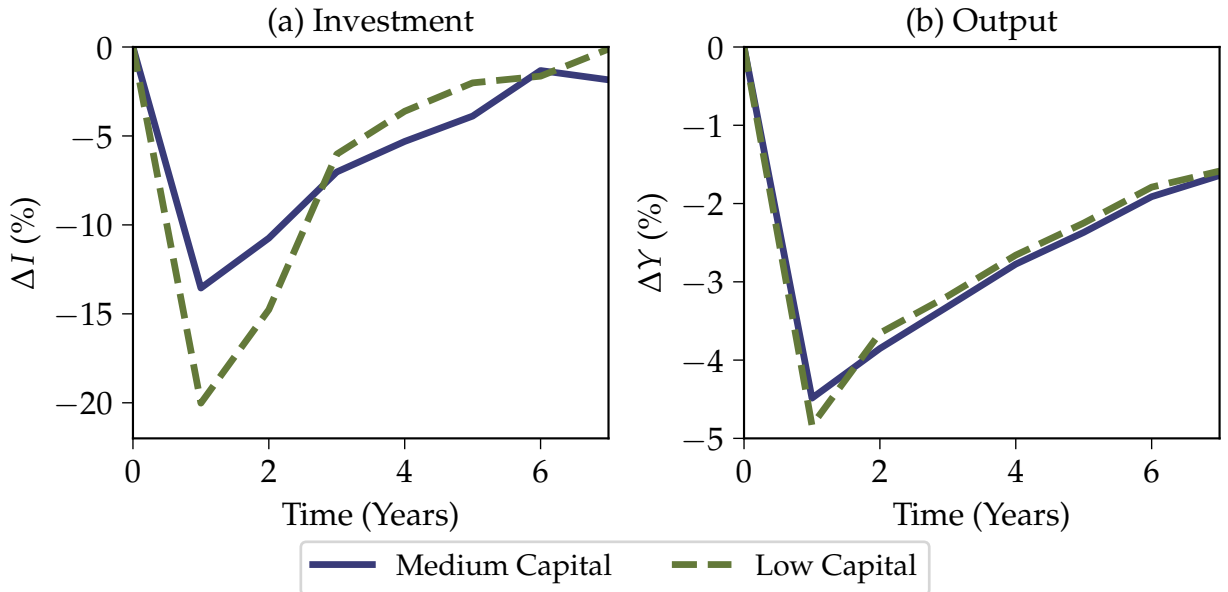


Note: The left panel plots the time series of aggregate investment across the three model variants; the baseline model (solid blue line), the model with financial friction only (orange dashed line) and the model with real frictions only (brown dotted line). The right panel plots the aggregate investment points across the different models (on the y-axis) relative to the aggregate investment points in the model with only real frictions. Again, the blue dots represent the baseline model, the orange triangles the model with financial frictions only and the dotted brown 45 degree line represents the model with only real frictions.

lumpiness, which propagates to the aggregate level and generates the observed nonlinear investment dynamics.

Overall, these results confirm that the lumpy investment spiral identified in Section 3 survives aggregation and meaningfully shapes business-cycle fluctuations. While prior work (e.g., Khan & Thomas (2008), Khan & Thomas (2013)) has shown that either real or financial frictions alone yield essentially linear aggregate dynamics—where firm heterogeneity is dampened by general equilibrium effects—we show that their interaction is key to producing nonlinear cycles. Supporting evidence from Table C3 in Appendix C.3 and Figure C3 in Appendix C.4 illustrates why: relative to the real-friction-only model, the baseline model features longer inaction spells and larger investment spikes, consistent with the theoretical prediction that financial frictions amplify lumpiness. Taken together, these findings demonstrate that the lumpy investment spiral strengthens micro-level nonlinearities and allows them to persist through aggregation, ultimately shaping the macroeconomic response to shocks.

Figure 5: State-dependent impulse response functions

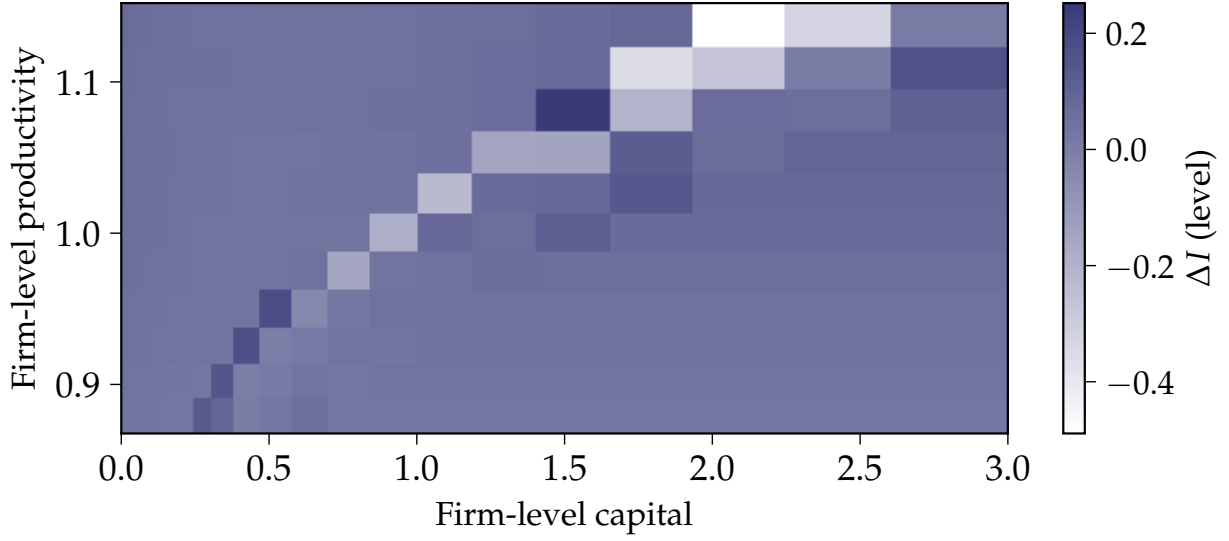


Note: Black solid line is the investment GIRF to a negative TFP shock when firm's internal resources are the median level over the cycle, while dashed red line is when resources are at the lowest point over the cycle. The dashed green line represents the GIRF under no financial frictions, in which case no state dependence happens, and so firms having median or low resources results in the same GIRF.

Generalized Impulse Response Functions (GIRFs). To better understand the role of the micro nonlinearities in shaping aggregate dynamics, we analyze the GIRF of investment and output following a negative TFP shock under two distinct scenarios: (i) when firms' internal resources are at their median level, and (ii) when they are at their lowest observed level across the entire simulation, which is 9% lower than the median. This comparison captures how financial slack conditions shape firms' reactions to shocks. When internal resources are scarce, firms are more likely to become trapped in the investment spiral, delaying investment and thereby deepening and prolonging the recession.

Figure 5 illustrates the GIRFs for both scenarios—median and low internal resources. When resources are low, investment (left panel) falls by 48% more relative to the median scenario. This amplification arises because more firms are drawn into the spiral: the lower the financial slack, the greater the fraction of firms postponing investment, resulting in a sharper aggregate contraction. This, in turn, translates into nonlinear output dynamics (right panel). Output declines by 8% more on impact when internal resources are low. The nonlinearity in output is less pronounced than in investment and dissipates faster, reflecting offsetting labor supply responses. Since capital is predetermined on impact, the short-run output adjustment stems from labor dynamics. As households anticipate

Figure 6: Differences in investment response in low and median k states



Note: X-axis capital, increasing from right to left. Y-axis productivity, increasing from right to left. Z-axis difference-in-differences measure of firm-level investment responses across states of internal resources, in the period when the shock happens, defined as $(I_{shock}^{LK} - I_{no\ shock}^{LK}) - (I_{shock}^{MK} - I_{no\ shock}^{MK})$, where *LK* and *MK* denote the low and median internal-capital states, respectively.

lower future capital due to the stronger investment drop, they reduce labor less sharply, cushioning part of the output decline.⁸

Figure 6 illustrates the micro-level mechanisms underlying the state dependence. The figure plots on the z-axis a difference-in-differences measure of firm-level investment responses across states of internal resources, in the period when the shock happens, formally defined as

$$(I_{shock}^{LK} - I_{no\ shock}^{LK}) - (I_{shock}^{MK} - I_{no\ shock}^{MK}),$$

where *LK* and *MK* denote the low and median internal-capital states, respectively. The x-axis represents firm-level capital and the y-axis represents firm-level productivity. The figure shows that the largest differences between the two states of the world arise among the most productive and capital-rich firms. In contrast, small and less productive firms are similarly affected in both states, implying that the drivers of state dependence are the largest and most productive firms.

Revisiting the theoretical model in Section 3 and Figure 3 helps clarify the mechanisms

⁸Notice that when we shut down either the financial friction or the real friction, no state dependence arises, as the investment or output GIRF are always the same no matter the firms' internal resources.

at work. The most productive firms have the highest marginal benefits of investment and typically rely more on external financing. Following a negative TFP shock, the marginal benefit of investment declines. Taking the derivative of the marginal benefit first with respect to TFP and then with respect to future capital shows that this decline is amplified when internal resources are low. The amplification is particularly pronounced for highly productive firms, for whom the marginal benefit curve is most affected when internal liquidity is scarce. This interaction causes the most productive firms to be the ones that fall into the investment spiral when internal resources are depleted—the key driver of the aggregate state dependence.

In addition, there is a distributional effect. When internal resources are low, a larger share of firms is already close to the inaction and spiral regions. A negative shock to productivity further compresses profits, pushing a greater fraction of firms into the spiral compared to periods when internal resources are abundant. This shift in the distribution of active versus inactive firms amplifies aggregate nonlinearity, as the mass of firms responding at the extensive margin increases precisely when the economy is in a weak state.

5.2 Endogenous financial disasters

So far, we have shown that TFP-driven business cycles with real and financial frictions can generate endogenous nonlinear dynamics. We now extend the analysis to additionally include financial shocks and show how this can give rise to endogenous disaster risk (Petrosky-Nadeau et al., 2018).

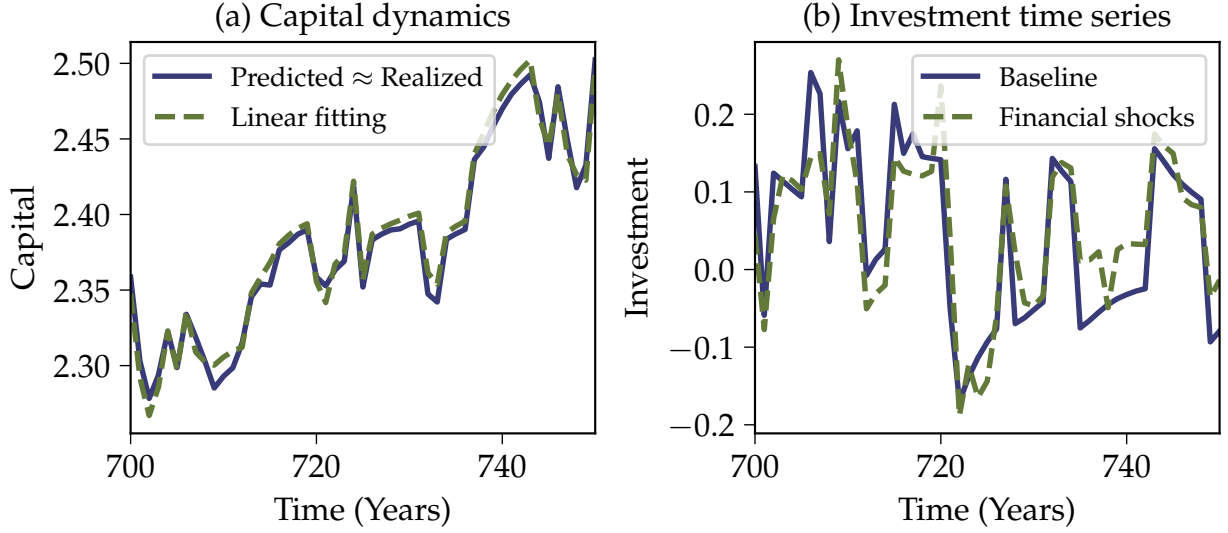
In addition to the TFP shock introduced in Section 4, we introduce an aggregate financial shock, allowing e_0 to fluctuate over time according to the stochastic process

$$e'_0 - e_0^{ss} = \rho_{e_0}(e_0 - e_0^{ss}) + \epsilon'_0, \quad \text{where } \epsilon'_0 \sim N(0, \sigma_{e_0}). \quad (37)$$

Parameters ρ_{e_0} and σ_{e_0} are set to 0.72 and 0.56, respectively, estimated from fitting the Gilchrist & Zakrajšek (2012) credit spread index to an AR(1) process. All other model parameters remain unchanged.

Figure 7 illustrates the resulting aggregate dynamics. Capital dynamics (left panel) display even stronger nonlinearities than in the baseline case, with the linear approximation performing particularly poorly. At the same time, investment (right panel) deviates sharply from the baseline, producing episodes of deep recessions. These downturns occur precisely when a negative financial shock coincides with periods of low internal resources, pushing many firms into prolonged investment inaction.

Figure 7: Aggregate fluctuations in the extended model with aggregate financial shock



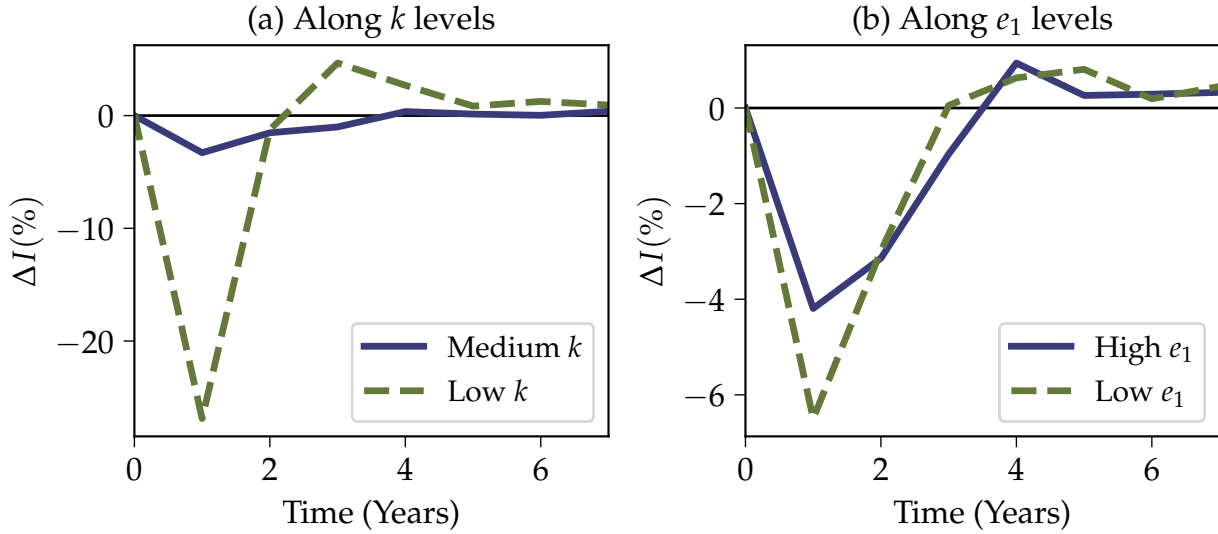
Note: On the left panel the time-series of capital dynamics. Black dotted line presents the linear fitting, the red dashed line the realized values and the solid blue line the predict values, which overlaps with the red line. On the right panel, the comparison of the investment times series in the baseline model (solid blue line) and the baseline model with a financial shock (red dashed line).

To further highlight this mechanism, we compute GIRFs to a financial shock under two conditions: when firms' internal resources are at their median value and when they are at their lowest observed level over the cycle. The left panel of Figure 8 shows that financial shocks are far more costly when firms are financially fragile, as limited internal resources amplify the investment collapse.

The mechanism behind this amplification can be understood by revisiting Figure 3. Following a TFP shock, firms' internal resources decline, increasing the likelihood of entering the investment spiral. However, the TFP shock also reduces the marginal benefit of investment, creating a counteracting force that tempers the number of firms moving into the spiral. A financial shock, by contrast, directly increases the wedge between internal and external financing costs without lowering the incentive to invest. As a result, more firms are pushed into inactivity, and no offsetting force mitigates the spiral.

The parameter e_1 further governs the strength of this mechanism. When e_1 is low, the external financing cost curve is relatively flat. In this case, a shift in e_0 has large aggregate consequences, as many firms that previously relied on external funds are forced into inaction. Conversely, when e_1 is high, the financing cost curve is steep: firms already use little external financing, so a shock to e_0 has smaller aggregate effects. The right panel of Figure 8 illustrates the investment response to a financial shock for average capital levels, when e_1 is 1% above or below the baseline value. When e_1 is higher, the aggregate

Figure 8: State-dependent investment impulse response functions: e_0 shock



Note: On the left panel the investment GIRF to a one standard deviation increase in e_0 when internal resources are at the median level (solid black line) and when they are at the lowest point over the cycle (dashed orange line). On the right panel the investment GIRF when internal resources are at the median level, when e_1 is one percent below (solid black line) or one percent above (dashed orange line).

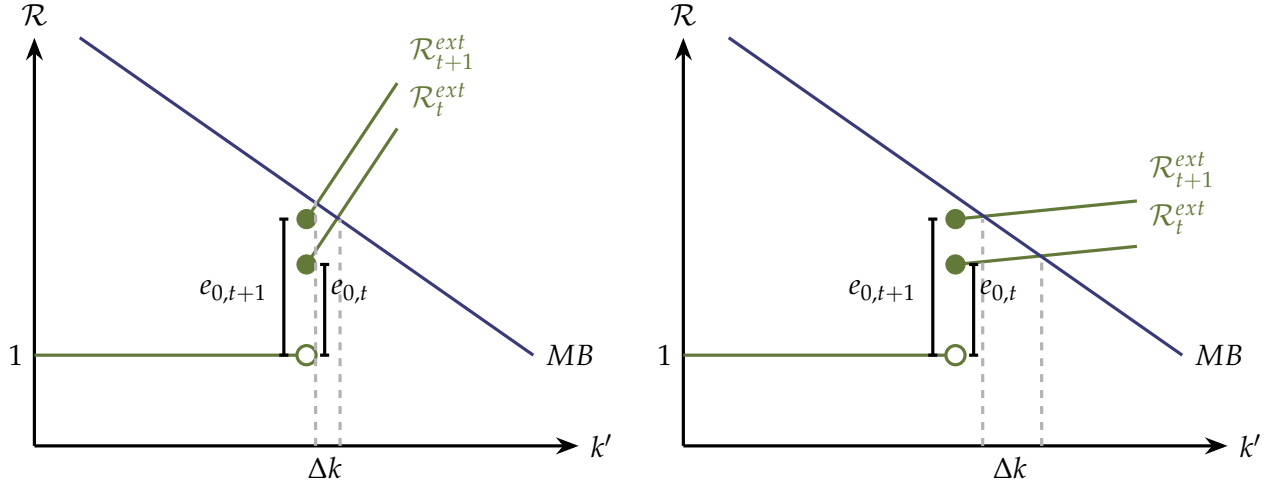
investment drops by 55% less than when e_1 is low.

Finally, Figure 9 illustrates this dependence graphically. With a high e_1 (left panel), external financing costs rise steeply with the size of the investment, leading firms to rely minimally on external funds, so an e_0 shock generates only small changes in k' . With a low e_1 (right panel), the flatter financing curve induces greater reliance on external finance, and the same e_0 shock produces much larger declines in investment.

Welfare Costs of Business Cycles. Finally, we assess the welfare implications of the nonlinear cycle. Because the lumpy investment spiral induces state-dependent downturns, and the presence of a financial shock creates endogenous disaster risks, the ergodic mean of consumption falls below its steady-state levels. This downward bias in long-run aggregates implies sizable welfare losses.

Table 2 quantifies these effects. It reports the ergodic mean and the 95% confidence interval of the ergodic mean as well as the steady state values of consumption for the baseline model and the extension with a financial shock. For the baseline model with a financial shock, the ergodic mean of consumption is statistically below their steady-state values. By contrast, when the financial shock is removed, the ergodic mean is significantly higher than the steady states, despite the model still being nonlinear. In fact, the nonlinearity of the model induces a stronger precautionary savings motive, which leads

Figure 9: Investment response to e_0 shock dependent on e_1



(a) High e_1 implies low investment variation

(b) Low e_1 implies high investment variation

Note: Both panels depict the effect of an equisizeable increase in e_0 conditional on a steep (a) and a shallow (b) slope of the marginal external financing cost.

Table 2: Welfare costs of the business cycle

Consumption	With fin. shock	Baseline
Ergodic mean	0.4157	0.4175
(95-CI)	(0.4150 0.4163)	(0.4165 0.4184)
Steady state	0.4164	0.4164
Ergodic mean < Steady state?	Yes	No

Note: The table reports the ergodic mean of consumption, the 95th confidence interval of consumption over the business cycle and the steady state value of consumption for both the baseline model with and without the financial shock.

to higher consumption overall. This highlights that the nonlinearity of the business cycle is not strong enough to by itself generate welfare costs over the business cycle. One needs endogenous disaster risks, which create deep, long-lasting and costly recessions, to have consumption falling below its steady-state levels.

6 Mechanism validation

Our theory implies that both in general equilibrium and even in a simple model, financial and real frictions reinforce each other. We have already provided some motivating stylized facts that this is the case - large investment activity is related to high debt issuance on the aggregate, and the average financial position of a firm is also negatively correlated to its average lumpiness of investment.

We now go further, and provide some more structural and causal evidence in favor of our mechanism. Firstly, we provide evidence that an arguably exogenous improvement in the financial position of a firm increases the likelihood of an investment spike. This is in line with our mechanism - as external financial costs fall, represented by e_0 and e_1 in our model, we should see a higher likelihood of spikes, *ceteris paribus*. We implement this notion empirically by broadly following the methodology outlined by [Alfaro et al. \(2024\)](#). We construct instruments by exploiting firms' differential exposure to monetary policy shocks to identify exogenous variation in firms' external financing cost. In particular, we exploit the fact that certain firms are more positively exposed to contractionary monetary policy shocks (for instance in certain industries or with low leverage) compared to other industries that have negative exposure to these shocks. We use daily

Table 3: Spike–Interaction Slopes for Change in Interest Expenses

	Leverage	Distance to Default	Liquidity
$\hat{\gamma}$	0.458** (0.225)	-0.036*** (0.009)	-0.653** (0.294)

Note. Each column corresponds to a separate regression of tomorrow's interest expense change (scaled by total liabilities) on the investment spike indicator interacted with one balance-sheet variable. All specifications control for contemporaneous Δ interest expense, firm size, firm age, cash flow, and sales growth, and include firm fixed effects together with industry–year fixed effects. The reported coefficient $\hat{\gamma}$ is the slope on $\text{Spike} \times X$, where X denotes leverage, distance-to-default, or liquidity, respectively. Standard errors, in parentheses, are two-way clustered by firm and year.

Larger investment spikes are associated disproportionate financial tightening Our second part of the empirical validation of our mechanism goes from investment spikes to the financial consequences. Our spiral mechanism implies that larger investment spikes - in terms of investment rate will imply a larger increase in the financing cost, as our schedule of e_1 and e_0 implies. As there is no direct measure of spread available in the Compustat file we proxy this again with the development of other financial variables after a investment spike. In order to understand whether larger investment rates are indeed associated

with larger deterioration of financing cost we run the following regression with

$$\begin{aligned} \Delta y_{jt+h} = & \beta_1 \left(i_{jt} | \text{spike}_{jt} \right) + \beta_2 \left(i_{jt} | \text{spike}_{jt} \right)^2 + \beta_3 \left(i_{jt} | \text{spike}_{jt} \right)^3 \\ & + \alpha_j + \alpha_{st} + \Gamma' X_{jt-1} + \varepsilon_{jt+h} \end{aligned} \quad (38)$$

7 Conclusion

This paper has documented and modeled a novel mechanism linking firms' investment lumpiness to financial frictions. Using firm-level U.S. data, we show that investment and financing decisions are tightly intertwined: the extensive margins of debt issuance and investment move together over time, and firms with lumpier investment patterns are more financially constrained. These empirical patterns motivate our central hypothesis—that large investment events are associated with higher financing costs.

We setup a two-period firm investment model, with both financial and real frictions, which illustrates that financial frictions alone can generate investment inaction, even in the absence of real adjustment costs. When the marginal benefit of investment lies between the cost of internal and external funds, firms with insufficient internal resources optimally postpone investment. The interaction between real and financial frictions amplifies this mechanism: postponement erodes internal resources, raising external financing costs and reinforcing inaction. We term this self-reinforcing dynamic the lumpy investment spiral.

Embedding this mechanism in a general equilibrium model shows that it survives aggregation and materially shapes business cycle dynamics. Financial frictions magnify investment lumpiness, prolong inaction spells, and induce strong state dependence in macroeconomic responses. When internal resources are scarce, negative productivity shocks trigger larger and more persistent declines in investment and output, as more firms are pushed into the spiral. Introducing exogenous financial shocks further reveals that this interaction can generate endogenous disaster risk—deep recessions that arise endogenously when low internal resources meet tightening external finance.

Empirical tests using Compustat data corroborate the model's predictions. Higher external financing costs, instrumented by monetary policy shocks, lead to fewer investment spikes and longer inaction spells. Conversely, investment spikes are followed by increases in proxies for financing spreads, consistent with the feedback mechanism implied by the model.

Taken together, these results highlight the importance of modeling jointly real and

financial frictions to understand the nonlinear nature of firm investment and its macroeconomic consequences. The lumpy investment spiral provides a micro-founded channel through which financial conditions shape the persistence and amplitude of business cycles.

References

- Alfaro, I., Bloom, N., & Lin, X. (2024). The finance uncertainty multiplier. *Journal of Political Economy*, 132(2), 577–615.
- Bachmann, R., Caballero, R. J., & Engel, E. M. R. A. (2013). Aggregate Implications of Lumpy Investment: New Evidence and a DSGE Model. *American Economic Journal: Macroeconomics*, 5(4), 29–67.
- Bachmann, R. & Ma, L. (2016). Lumpy investment, lumpy inventories. *Journal of Money, Credit and Banking*, 48(5), 821–855.
- Baley, I. & Blanco, A. (2021). Aggregate dynamics in lumpy economies. *Econometrica*, 89(3), 1235–1264.
- Baley, I. & Blanco, A. (2024). *The Macroeconomics of Irreversibility*. FRB Atlanta Working Paper 2024-17, Federal Reserve Bank of Atlanta.
- Bazdresch, S. (2013). The role of non-convex costs in firms' investment and financial dynamics. *Journal of Economic Dynamics and Control*, 37(5), 929–950.
- Begenau, J. & Salomao, J. (2018). Firm Financing over the Business Cycle. *The Review of Financial Studies*, 32(4), 1235–1274.
- Bernanke, B. S., Gertler, M., & Gilchrist, S. (1999). The financial accelerator in a quantitative business cycle framework. *Handbook of macroeconomics*, 1, 1341–1393.
- Caballero, R. J. & Engel, E. M. R. A. (1999). Explaining investment dynamics in u.s. manufacturing: A generalized (s, s) approach. *Econometrica*, 67(4), 783–826.
- Cloyne, J., Ferreira, C., Froemel, M., & Surico, P. (2023). Monetary Policy, Corporate Finance, and Investment. *Journal of the European Economic Association*. jvad009.
- Cooper, R., Haltiwanger, J., & Power, L. (1999). Machine replacement and the business cycle: Lumps and bumps. *American Economic Review*, 89(4), 921–946.
- Cooper, R. W. & Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *The Review of Economic Studies*, 73(3), 611–633.
- Crouzet, N. & Mehrotra, N. R. (2020). Small and large firms over the business cycle. *American Economic Review*, 110(11), 3549–3601.
- Ferreira, M., Haber, T., & Rorig, C. (2023). *Financial Constraints and Firm Size: Micro-Evidence and Aggregate Implications*. Working Papers 777, DNB.
- Ferreira, T. R. T., Ostry, D. A., & Rogers, J. (2024). *Firm financial conditions and the transmission of monetary policy*. Bank of England working papers 1093, Bank of England.
- Gilchrist, S. & Zakrajšek, E. (2012). Credit spreads and business cycle fluctuations. *American economic review*, 102(4), 1692–1720.

- Gilchrist, S. & Zakrajšek, E. (2012). Credit spreads and business cycle fluctuations. *American Economic Review*, 102(4), 1692–1720.
- Gnewuch, M. & Zhang, D. (2025). Monetary policy, firm heterogeneity, and the distribution of investment rates. *Journal of Monetary Economics*, 149, 103721.
- Gourio, F. & Kashyap, A. K. (2007). Investment spikes: New facts and a general equilibrium exploration. *Journal of Monetary Economics*, 54, 1–22.
- Hennessy, C. A. & Whited, T. M. (2007). How costly is external financing? evidence from a structural estimation. *The Journal of Finance*, 62(4), 1705–1745.
- Jeenas, P. (2023). *Firm Balance Sheet Liquidity, Monetary Policy Shocks, and Investment Dynamics*. Working Papers 1409, Barcelona School of Economics.
- Jermann, U. & Quadrini, V. (2012). Macroeconomic effects of financial shocks. *American Economic Review*, 102(1), 238–271.
- Jiao, F. & Zhang, C. (2022). Lumpy investment and credit risk. *Journal of Corporate Finance*, 77, 102293.
- Khan, A. & Thomas, J. K. (2008). Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate Investment Dynamics. *Econometrica*, 76(2), 395–436.
- Khan, A. & Thomas, J. K. (2013). Credit shocks and aggregate fluctuations in an economy with production heterogeneity. *Journal of Political Economy*, 121(6), 1055–1107.
- Kiyotaki, N. & Moore, J. (1997). Credit cycles. *Journal of political economy*, 105(2), 211–248.
- Koby, Y. & Wolf, C. (2020). Aggregation in heterogeneous-firm models: Theory and measurement. Manuscript.
- Lee, H. (2022). Striking While the Iron Is Cold: Fragility after a Surge of Lumpy Investments. *Working Paper*, (pp.68).
- Lee, H. (2025). Global nonlinear solutions in sequence space and the generalized transition function. *Working paper*.
- Melcangi, D. (2024). Firms’ precautionary savings and employment during a credit crisis. *American Economic Journal: Macroeconomics*, 16(1), 356–386.
- Merton, R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *The Journal of Finance*, 29(2), 449–470.
- Ottonello, P. & Winberry, T. (2020). Financial heterogeneity and the investment channel of monetary policy. *Econometrica*, 88(6), 2473–2502.
- Ottonello, P. & Winberry, T. (2024). *Capital, Ideas, and the Costs of Financial Frictions*. Working Paper 32056, National Bureau of Economic Research.
- Petrosky-Nadeau, N., Zhang, L., & Kuehn, L.-A. (2018). Endogenous disasters. *American*

- Economic Review*, 108(8), 2212–45.
- Senga, T., Thomas, J., & Khan, A. (2017). *Default Risk and Aggregate Fluctuations in an Economy with Production Heterogeneity*. 2017 Meeting Papers 889, Society for Economic Dynamics.
- Thomas, J. K. (2002). Is lumpy investment relevant for the business cycle? *Journal of political Economy*, 110(3), 508–534.
- Whited, T. M. (2006). External finance constraints and the intertemporal pattern of intermittent investment. *Journal of Financial Economics*, 81(3), 467–502.
- Winberry, T. (2021). Lumpy Investment, Business Cycles, and Stimulus Policy. *American Economic Review*, 111(1), 364–396.
- Zwick, E. & Mahon, J. (2017). Tax policy and heterogeneous investment behavior. *American Economic Review*, 107(1), 217–48.